

| Assessment | Test | Domain        | Skill               | Difficulty                                        |
|------------|------|---------------|---------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Easy <div><div></div><div></div><div></div></div> |

Question

The function  $f$  is defined by  $f(x) = x^3 + 9$ . What is the value of  $f(2)$ ?

Answer

- A. 14
- B. 15
- C. 17
- D. 18

Correct Answer: C

Rationale

Choice C is correct. It's given that  $f(x) = x^3 + 9$ . Substituting 2 for  $x$  in this equation yields  $f(2) = (2)^3 + 9$ . This is equivalent to  $f(2) = 8 + 9$ , or  $f(2) = 17$ .

Choice A is incorrect. This is the value of  $2 + 3 + 9$ , not  $2^3 + 9$ .

Choice B is incorrect. This is the value of  $2(3) + 9$ , not  $2^3 + 9$ .

Choice D is incorrect. This is the value of  $3^2 + 9$ , not  $2^3 + 9$ .